



BIRZEIT UNIVERSITY

Faculty of Engineering and Technology

Department of Computer Science

COMP333—DATA BASE SYSTEMS

Phase 1: Requirements Document

Prepared by:

Yahya Hasan 1242481 Section 5

Mohammed Quraan 1243436 Section 5

Instructor: Yousef Hassouneh

Date: April 22, 2026

Contents

1	Group Information	2
2	Project Title	2
3	Project Scope and Description	2
4	Client Information	3
4.1	Customer Segments	3
4.2	Key Entities	3
5	Technology	4
6	Sample Queries	4
6.1	Retrieval Queries	4
6.2	Insertion Queries	6
6.3	Modification Queries	7

1 Group Information

Group Name: Bawabet Al-Quds Roastery

Group Members:

- Yahya Hasan — ID: 1242481
- Mohammed Quraan — ID: 1243436

2 Project Title

Bawabet Al-Quds Roastery: An Integrated Customer and Admin Management System for Sales and Orders

3 Project Scope and Description

This project builds a database-driven application for **Bawabet Al-Quds Roastery**, a specialty coffee roastery and distribution business based in Ramallah, Palestine. Two types of users interact with the system: **customers**, who browse and buy products, and an **admin**, who handles the catalog, packages, and orders.

- **Customer-Facing Shopping:** Customers browse coffee, spices, and herbs, and can filter, sort, or search the catalog by name. They add items to a cart and review a running total before placing the order.
- **Order Management:** Once a customer checks out, the cart becomes an order. Customers can see their own history, while the admin sees all orders across every customer.
- **Admin Catalog Control:** The admin manages products and can also put together *packages* — grouped items sold as one unit, usually at a discounted price.
- **Warehouse and Inventory:** Stock levels will be tracked in the database. We plan to expand the warehouse side in a later phase once requirements are clearer.

The database will cover the following functional areas:

- Product catalog management (coffee, spices, and herbs) with filtering, sorting, and name-based search
- Package creation and management by the admin
- Customer account management (Individual and Wholesale/B2B)
- Shopping cart and order processing
- Order history per customer and full order overview for the admin
- Branch management and per-branch sales tracking
- Supplier management and supply records per branch and product
- Payment processing and employee-payment linkage
- Warehouse and inventory tracking per branch

4 Client Information

Client Name: Bawabet Al-Quds (Jerusalem Gate) Roastery

Business Type: Retail and Wholesale Distribution of Specialty Coffee, Spices, and Herbs

Location: Ramallah, Palestine

4.1 Customer Segments

The business deals with two kinds of customers:

- **Individual Customers:** Regular buyers who order for personal use at standard retail prices.
- **Wholesale and B2B Clients:** Supermarkets, grocers, and distributors who order in bulk and resell. They typically get different pricing and their orders are tracked at the account level.

4.2 Key Entities

The main things we are tracking in the database are:

- **Customer:** Anyone with an account, either individual or wholesale, with their contact details and order history stored.
- **Admin:** Has full control over the product catalog and packages, and can see all orders on the platform.
- **Product:** Anything sold in the store, split into Coffee, Spices, or Herbs. Each product has a name, description, price, and a stock count.
- **Package:** A bundle put together by the admin and sold as one item, like a gift set or a promotional deal.
- **Cart:** Holds what the customer has picked before they finalize the purchase.
- **Order:** A finalized purchase — what was bought, how much, by whom, and when.
- **Branch:** A physical store location of the roastery. Each branch maintains its own inventory, handles its own sales transactions, and is staffed by its own employees.
- **Employee:** A staff member assigned to a branch who processes customer payments and handles sales transactions.
- **Supplier:** An external vendor who supplies raw or finished products to one or more branches. Each supply record captures the product, the branch, the quantity, and the supply date.
- **Payment:** A financial transaction that settles a customer order, capturing the amount paid, payment method, date, and the employee who processed it.
- **Warehouse Inventory:** Keeps track of how much of each product is in stock at each branch.

5 Technology

- **Application Layer:** Java (using JavaFX & CSS for the graphical user interface and JDBC for database connectivity).
- **Database Management System:** MySQL — it is what we have been working with in the course and it handles the queries we need without issues.

6 Sample Queries

The following queries demonstrate the intended functionality of the database application for Bawabet Al-Quds Roastery. They are grouped by operation type: retrieval (SELECT), insertion (INSERT), and modification (UPDATE/DELETE).

6.1 Retrieval Queries

- Q1.** Retrieve all products in a specified category (Coffee, Spices, or Herbs), sorted by product name.
- Q2.** Retrieve all products whose name contains a given search keyword, sorted by category then by name.
- Q3.** Retrieve all products filtered by category and sorted by price (ascending or descending) as chosen by the user.
- Q4.** Retrieve all products filtered by category and sorted by name (A–Z or Z–A) as chosen by the user.
- Q5.** Retrieve the full details of a specific product given its ID, including name, category, description, price, and current stock quantity.
- Q6.** Retrieve all packages created by the admin, including each package’s name, description, total price, and the list of products it contains.
- Q7.** Retrieve the full details of a specific package, including all contained products with their individual prices and quantities.
- Q8.** Retrieve the contents of a specific customer’s active cart, showing each item’s name, quantity, unit price, and line total, along with the cart grand total.
- Q9.** Retrieve the number of distinct items and the total quantity of units currently in a specific customer’s cart.
- Q10.** Retrieve all orders placed by a specific customer, sorted by order date descending.
- Q11.** Retrieve the full details of a specific order, including the customer name, order date, all products and packages ordered with their quantities and unit prices, and the order total.
- Q12.** Retrieve all orders placed across all customers within a specified date range, sorted by order date descending (admin view).
- Q13.** Retrieve all pending orders (not yet fulfilled) across all customers, sorted by order date ascending (admin view).

- Q14.** Retrieve the total number of orders and total revenue grouped by customer type (Individual vs. Wholesale/B2B) for a specified period.
- Q15.** Retrieve the top 5 best-selling products by total quantity ordered during a specified period, sorted by total quantity descending.
- Q16.** Retrieve the top 3 most ordered packages during a specified period, sorted by order count descending.
- Q17.** Retrieve all products that have been included in at least one package, along with the names of the packages they belong to.
- Q18.** Retrieve all products that are currently not part of any package.
- Q19.** Retrieve all orders that contain a specific product, including the customer name, quantity ordered, and order date, sorted by order date descending.
- Q20.** Retrieve all wholesale/B2B customers along with the total value of their orders, sorted by total order value descending.
- Q21.** Retrieve the total revenue generated per product category over a specified period, sorted by revenue descending.
- Q22.** Retrieve all products whose current stock quantity is below a specified threshold, sorted by stock quantity ascending.
- Q23.** Retrieve the full order history for a specific product, showing the customer name, quantity ordered, and order date for each order, sorted by date descending.
- Q24.** Retrieve all customers who have placed at least one order, along with the count of their orders and the sum of their total spending, sorted by total spending descending.
- Q25.** Retrieve all packages that contain at least one product from a specified category, sorted by package name.
- Q26.** Retrieve the 10 most recently added products to the catalog, sorted by creation date descending.
- Q27.** Retrieve all customers who have not placed any order yet.
- Q28.** Retrieve the average order value grouped by customer type (Individual vs. Wholesale/B2B).
- Q29.** Retrieve the total quantity sold per product over all time, including products with zero sales, sorted by total quantity sold descending.
- Q30.** List all customers who purchased a specific product from a specified branch, showing customer name, quantity bought, order date, and branch name, sorted by order date descending.
- Q31.** Retrieve the most-selling product category across all branches within a specified time period, based on total quantity sold per category.
- Q32.** Retrieve the branch that generated the highest total sales amount (revenue) in a specified month, showing branch name and total revenue.

- Q33.** List all payment details for a specific customer, including payment date, amount, payment method, order ID, and the name of the employee who processed each payment, sorted by payment date descending.
- Q34.** List all suppliers who supply a specific product exclusively to a single specified branch (i.e., they do not supply that product to any other branch), showing supplier name and contact details.
- Q35.** Retrieve the total quantity of each product supplied by each supplier within a specified date range, sorted by supplier name then product name (aggregation: **SUM** of supplied quantities).
- Q36.** Retrieve all suppliers for a specified product category, showing supplier name, contact information, and the products they supply in that category.
- Q37.** Retrieve the average supply quantity per supplier across all branches and the total number of supply transactions they have made, sorted by average quantity descending (**AVG**, **COUNT** aggregation).
- Q38.** Retrieve all products whose price falls within a specified range (*min_price–max_price*), sorted by price ascending.
- Q39.** Retrieve all orders whose total value falls within a specified amount range, showing customer name, order date, and order total, sorted by order total descending.
- Q40.** Retrieve total revenue, average order value, minimum order value, and maximum order value per branch over a specified period (**SUM**, **AVG**, **MIN**, **MAX** aggregation).
- Q41.** Retrieve the total number of orders and total revenue grouped by branch for a specified time period, sorted by total revenue descending (**COUNT**, **SUM** aggregation).
- Q42.** Retrieve the average number of items per order placed in a specified date range, grouped by branch (**AVG** aggregation).

6.2 Insertion Queries

- I1.** Insert a new customer account with name, contact information, and type (Individual or Wholesale/B2B).
- I2.** Insert a new product into the catalog with name, category, description, price, and initial stock quantity.
- I3.** Insert a new package with a name, description, price, and the set of products it contains along with their quantities.
- I4.** Insert a new item (product or package) into a specific customer's active cart with a given quantity.
- I5.** Insert a new order record when a customer confirms their cart, capturing the customer ID, order date, and a snapshot of all cart items with their quantities and prices at the time of purchase.
- I6.** Insert a new warehouse stock entry when a product is restocked, recording the product, quantity added, and date.

6.3 Modification Queries

- M1.** Update the price of a specific product.
- M2.** Update the stock quantity of a specific product after a restock or after an order is fulfilled.
- M3.** Update the description or name of an existing package.
- M4.** Update the quantity of a specific item in a customer's cart.
- M5.** Remove a specific item from a customer's cart.
- M6.** Clear all items from a customer's cart (e.g., after order confirmation or on explicit customer request).
- M7.** Update the status of a specific order (e.g., from *Pending* to *Fulfilled* or *Cancelled*).
- M8.** Remove a product from the catalog (only if it is not referenced by any existing order or active package).
- M9.** Remove a package from the catalog (only if it is not part of any active or pending order).
- M10.** Update a customer's contact information (phone number or address).